

Q1) What is GCC and how is it related to GNU?

- a. GCC is a compile command within the terminal, while GNU is a free software that houses the GCC command.

Q2) How does visual studio code use GCC on the raspberry PI for our labs?

- a. When using VS Code on the PI it automatically calls the GCC command when hitting run.

Q3) Consider the program shown below:

- a. Identify all of the directives in this program?
  - a. 2 directives: stdio & stdbool
- b. What output does the program produce?
  - a. The program outputs whether the inputted integer is or is not a prime number
- c. What input does the program use?
  - a. The program utilizes a integer input
- d. What are the names of the variables (identifiers) in this program?
  - a. The names of the identifies within the program are: is\_prime, n\_arg, divisor, main, and n
- e. What are the types of the variables in this program?
  - a. The two types of variables are integers and booleans
- f. How many statements are in this program?
  - a. 13
- g. What do the printf 's do in this program?
  - a. The printf outputs a formatted print statement to the terminal prompting the user to input a number and then printing out if that number is prime or not
- h. What does scanf do in this program?
  - a. The scanf reads what number the user inputs in to the terminal to then determine if that number is a prime or not
- i. What does the “\n” do in this program?
  - a. The “\n” creates a new line when printing to the terminal

3.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    float radius;
```

```
    float vol;
```

```
    printf("Enter the radius of a sphere: ");
```

```
    scanf("%f", &radius);
```

```
    vol = ((4.0f/3.0f) * (3.1415) * radius * radius * radius);
```

```
    printf("The volume of the sphere is: %f\n", vol);
```

```
    return 0;
```

```
}
```

5.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int xInt, ans;
```

```
    printf("Enter a value for X: ");
```

```
    scanf("%d", &xInt);
```

```
    ans = (3 * (xInt * xInt * xInt * xInt * xInt));
```

```
    ans += (2 * (xInt * xInt * xInt * xInt));
```

```
    ans -= (5 * (xInt * xInt * xInt));
```

```
    ans -= (xInt * xInt);
```

```
    ans += ((7 * xInt) - 6);
```

```
    printf("Answer: %d\n", ans);
```

```
}
```

7.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int dllrAmt;
```

```
    printf("Enter a dollar amount: ");
```

```
    scanf("%d", &dllrAmt);
```

```
    printf("\n$20 bills: %d\n", dllrAmt / 20);
```

```
    dllrAmt %= 20;
```

```
    printf("$10 bills: %d\n", dllrAmt / 10);
```

```
    dllrAmt %= 10;
```

```
    printf("$5 bills: %d\n", dllrAmt / 5);
```

```
    dllrAmt %= 5;
```

```
    printf("$1 bills: %d\n", dllrAmt / 1);
```

```
    return 0;
```

```
}
```

### Chapter 3, Questions 1, 4, & 5

1. a. 86, 1040

b.  $3.02530 \times 10^1$

c. 83.1620

d.  $1 \times 10^{-6}$

4. i. 10

x. 0.300000

j. 5

5. i. 12.300000

x. 45

j. 0.600000